BIOL3203 Food Microbiology

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2025-2026 Semester 1

Introduction

This course begins with a discussion of marine biology, covering the Introduction to Food Microbiology, Foodborne Pathogens, Detection and Enumeration of microbes in foods, Spore and their significant, Beneficial Bacteria, Factors that influence microbes in food and Physical Method of Food Preservation. This course aims to introduce the microscopic of food. Upon completion, students will be familiar with the mechanism and phenomenon in food microbiology.

Contents

1 Lecture 4 Detection and Enumeration of Microbes in Food III

1. Spiral Plate Count
   - Determination of bacterial numbers in liquid food samples
     ⇒ E.g. Milk
     ∵ Food particles may cause blocking of the dispenser
   - Robotic Arm : Dispense a sample in decreasing volume of the inoculum spirally
     ⇒ Direction : From the centre towards outside
     ⇒ Generate a concentration range of 1:10000
   - Centre Part has the highest concentration (Higher Density of Cells)
     ⇒ Fewer colonies at the edge of the plate
   - Specialized Counting Grid : Used to count colonies on a plate

Counting grid for Spiral plate

Figure 1: Counting Grid for Spiral Plate

2. Membrane Filtration
   - Allow Fluid Samples, Viscous Sample (Samll Amount) and Air Samples passes through the Membrane Filter to collect bacteria
     ⇒ E.g. Water, Diluent, Milk, Juice, Gas

- Membrane Filter : Cellulose / Polycarbonate Nucleopore (Pore Size : 0.45 mm)
- Can detect low number of microbes
  ∵ Large volumes (> 1 L) can be passed through a single filter
- Allows isolation and enumeration of discrete colonies of bacteria
- Reduce Preparation time
- Provides result within 24 hours (Incubation Time)

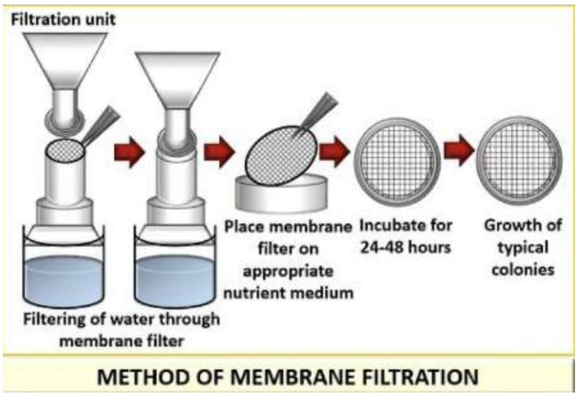

Figure 2: Membrane Filtration

3. Qualitative Culture Methods
   - Determine the presence of specific microorganisms in food
   - Include amplification, selection and confirmation
   - Detect viable organisms
4. Detection of ***Salmonella***
   - Guideline : Standardized by ISO 6579:2002 for ***Salmonella*** detection
   - This method will undergo Pre-enrichment by using Buffered Peptone Water (BPW)
   - Then Undergo Selective Enrichment
   - Selective Enrichment : Rappaport Vassialidis (RV) Broth, Tetrathionate (TT) Broth, Selenite Cystine (SC) Broth

- Then, Streak the enrichment sample on differential medium ⇒ E.g. Bismuth sulfite (BS), Hoektoen Enteric (HE), Xylose Lysine Deoxycholate (XLD)
- Colonies Appeared : Proceed for biochemical and Serological Detection
- Temperature : 35-43°C

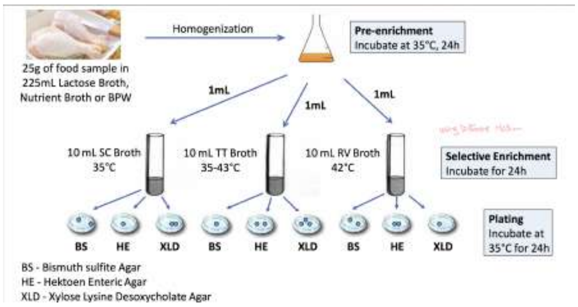

Figure 3: Qualitative Culture Methods Procedural

5. Rapid Method
   - Principles
     (a) Direct Counting of Microbes
     (b) Detection and measurement of growth end products.
     (c) Detection and measurement of growth or growth-related processes
     (d) Automate manipulative procedures
     (e) Reduce the number of steps
     (f) Reduce the scale of operation
   - Used to enumerate microorganisms in samples
   - Type of Rapid Methods
     (a) Direct Microscopic Count (DMC)
     (b) Physical Methods
       - Impedance
       - Flow Cytometry
       - Radiometry
       - Optical Density

(c) Metabolism Based Method
   - Sugar Fermentation
   - Nitrate Reduction
   - Fat Hydrolysis
   - Indole Production
   - ATP Bioluminescence

(d) Immunoassay
   - ELISA
   - RIA
   - DFA

6. Direct Microscopic Count (DMC)
   - Use Counting Chamber
     - Petroff-Hausser Counter : For Sperms and Bacteria
     - Superior Marienfeld Hemocytoameter : For any Microorganisms
     - Howard Mold Counting Chamber : For Molds in tomato products
   - Only use small amount of culture (10 μl)
   - Place the sample with cover slip
   - Capillary Action : Brings the liquid onto the viewing area with the grid (25 Large Square in 0.2 mm$\times$0.2 mm)
     $\Rightarrow$ Only count 5 squares per grid $\Rightarrow$ Only count 2 edges per square in one grid
   - Number of Bacteria : Directly counted microscopically
     $\Rightarrow$ Determined by Extrapolation

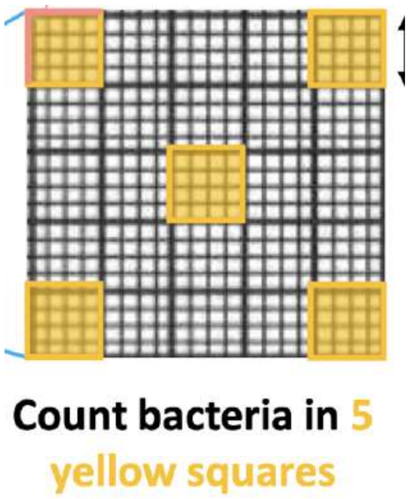

Figure 4: Bacteria Counting in DMC

- DMC Equation

$$\text{Bacteria Number} = \frac{\sum \text{Counts of Chambers}}{\text{Number of Square}} \times \frac{1}{\text{Square Volume}} \times \text{Dilution}$$

- SI Unit : /nl
  $\Rightarrow \times 10^6/\text{ml}$

7. Physical Method 1 : Impedance

- Impedance Microbiology : Tracing the Microbial Growth by measuring the change in electrical conductivity in growth medium
- Macronutrients are metabolized into **Small Charged Components**
  $\Rightarrow$ Support Microbial Growth + $\uparrow$ Electrical Conductivity
  $= \downarrow$ Impedance
  $\Rightarrow$ E.g. Lactic Acid, Carbon Dioxide, Ammonia

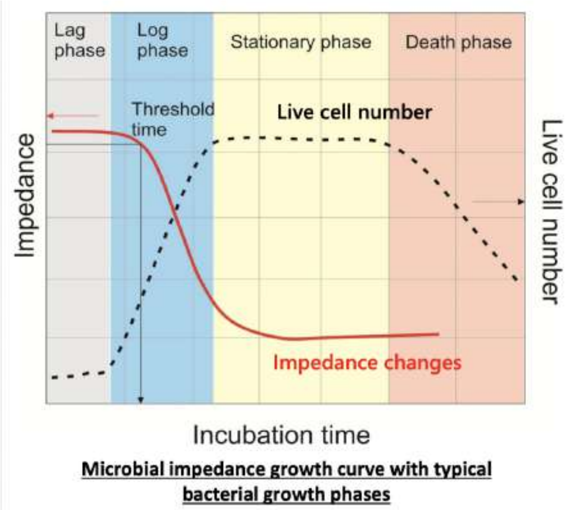

Figure 5: Microbial Impedance Growth Curve

8. Physical Method 2 : Flow Cytometry

- Flow Cytometry : Detect single or multiple microorganisms rapidly + Perform Live Cell Counting
- Flow through the apparatus in a fluid stream, Equipped with sorting capability
  ⇒ Allow Separation of cells

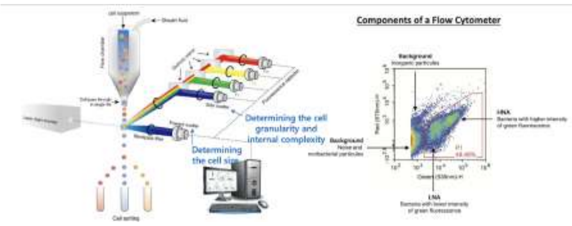

Figure 6: Flow Cytometry

9. Physical Method 3 : Radiometry

- Radiometric Detection : Based on the Incorporation of ^{14}C−Labelled Metabolite
  ⇒ E.g. $^{14}CO_2$ − Glucose, $^{14}CO_2$ − Formate, $^{14}CO_2$ − Glutamate
- When organisms utilize this labelled metabolite → $^{14}CO_2$ is released + Measured by Radiometer Counter

10. Physical Method 4 : Optical Density (OD)
    - Bacterial Growth in Culture Broth : Turning Turbid
    - Spectrophotometer : Measure the scattering of light passing through the media
    - $\uparrow$ Absorbance, $\uparrow$ Number of Bacteria

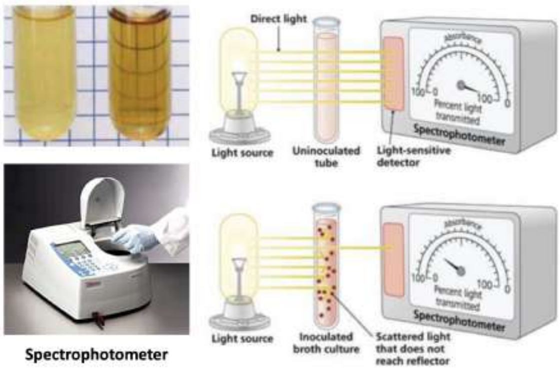

Figure 7: Optical Density (OD)

11. Metabolism Based Method 1 : Sugar Fermentation
    - Fermentative Degradation of Carbohydrates by microorganisms
      $\Rightarrow$ Necessary Condition : Anaerobic Condition
      $\Rightarrow$ E.g. Monosaccharides, Disaccharides, Polysaccharides
    - Fermentation Tube : Specific Carbohydrate + pH Indicator
      $\Rightarrow$ Example of pH Indicator : Phenol Red
    - Durham tube : Detect Gas Production
    - Organisms ferments carbohydrates $\rightarrow$ Produce organic acid $\rightarrow$ pH $\downarrow$ $\rightarrow$ Turn the medium into yellow colour
      $\because$ pH decreasing cause Colour Change
      $\Rightarrow$ Example of Organic Acid : Lactic Acid , Formic Acid, Acetic Acid

12. Metabolism Based Method 2 : Nitrate Reduction
    $\Rightarrow$ $NO_3^- \rightarrow N_2$ or Nitrogenous Medium
    - Gram-negative Bacteria use nitrate as Final Electron Acceptor in anaerobic metabolism

- Organism produce Nitrate Reductase → Reduce Nitrate to Nitrite → Reduce into Nitrogenous Gases
- Detect Nitrite : Use Sulfanilic Acid
  ⇒ React to nitrite to form a complex called **Nitrite-sulfanilic Acid**
  ⇒ **Nitrite-sulfanilic Acid** reacts with α-naphthylamine
  ⇒ Get a Red Precipitate (Water-soluble azo dye)
- Colour Change = Formation of NO_2

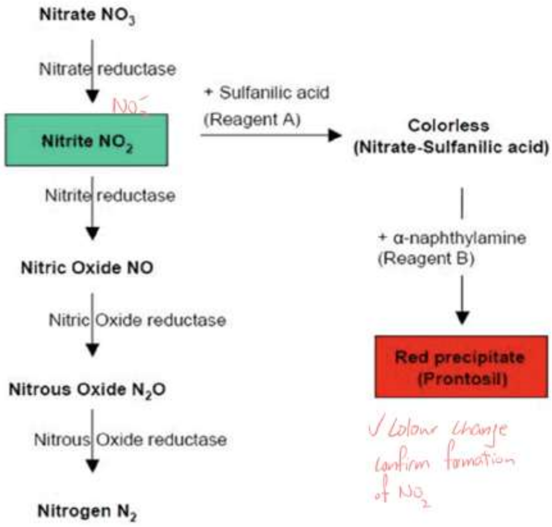

Figure 8: Nitrate Reduction Pathway

13. Zinc Powder

- 2nd follow up test for reduction of nitrate
  ⇒ Catalyses the reduction of nitrate to nitrite
- It will produce red colour = Nitrate is not reduced by organisms
- Organism Reduce nitrate = No Colour Change
  ⇒ We can say the organism is a nitrate reducer
- Opposite to Sulfanilic Acid

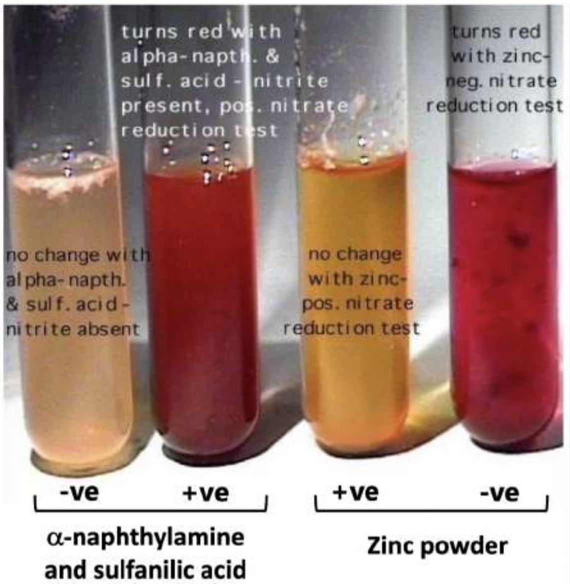

Figure 9: Sulfanilic Acid vs Zinc Powder

14. Metabolism Based Method 3 : Fat Hydrolysis

- Before Metabolism of Fat : Bacteria degrade triacylglycerol by Lipases with **hydrolysis**
- Tributyrin Agar Plate is used
- After Inoculation and Incubation : Organisms Excreting Lipase + Show a Zone of Lipolysis
  $\Rightarrow$ It is a clear area (Loss of Opacity = Transparency) surrounding the bacterial growth
- Transparency Zone : Hydrolytic Reaction yielding Soluble Glycerol and Fatty Acid
  $\Rightarrow$ Represents a Positive Reaction for lipid Hydrolysis
- If Lipolytic Enzyme is absent $\rightarrow$ Medium Remain Opacity = Negative Reaction

Figure 10: Fat Hydrolysis

15. Metabolism Based Method 4 : Indole Production Test

- Used to demonstrates the ability of Enterobacteria
  ⇒ Decompose Tryptophan in Reductive Deamination Process
  ⇒ Form Indole by Tryptophanase
  ⇒ E.g. ***E. coli*** is one of the indole

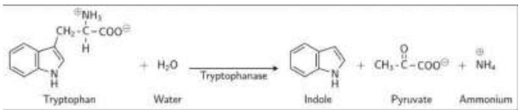

Figure 11: Indole Production Test

- Kovac's Reagent : Hydrochloric Acid + p-dimethylaminobenzaldehyde in amyl acohol (Not water-soluble)
- Indole + Kovac's Reagent → Solution have colour change
  ⇒ From Pungent Yellow to Cherry Red in an oily layer at the top of broth

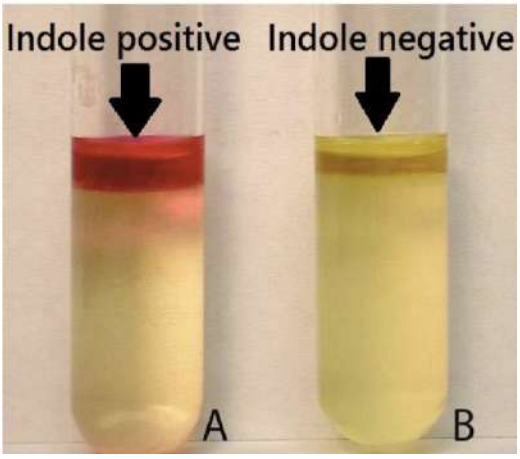

Figure 12: Colour Change of Indole Test

16. Metabolism Based Method 5 : ATP Bioluminescence

- Detect the amount of ATP
- A indirect measurement of the amount of organic substance and food residue on the surface
- It has Light Emission from **Luciferin-Luciferase Reaction**

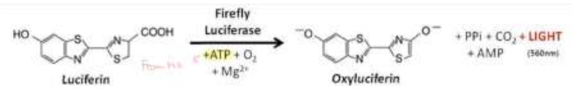

Figure 13: Luciferin-Luciferase Reaction

- Light Emission Measurement : Use Portable ATP Meters
  ⇒ Very Sensitive and Instantaneous

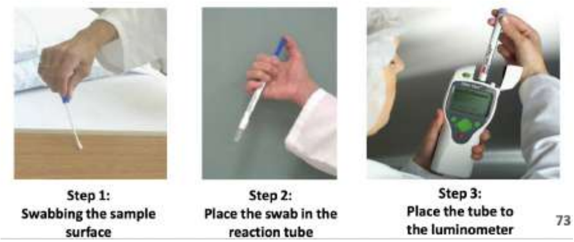

Figure 14: ATP Bioluminescence Steps

- Light Emission Unit : Relative Light Units (RLUs)

- ↑ RLU, ↑ ATP on the surface
- Limitation : This assay is non-specific method
  ⇒ Not able to discriminate between different microbial sources of contamination

17. Metabolism Based Method 6 : Analytical Profile Index (API)

- A commercial Strip with a panel of 20 Biochemical Tests
  ⇒ Tests Example : Sugar Fermentation, Amino Acid Utilization
- It can use for identification

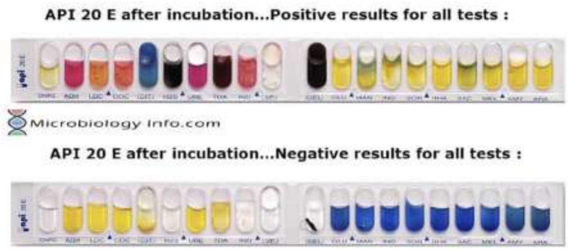

Figure 15: Analytical Profile Index

18. Immunoassay

- Immunological Methods : Base on Specific Binding between **Antigen** and **Antibody**

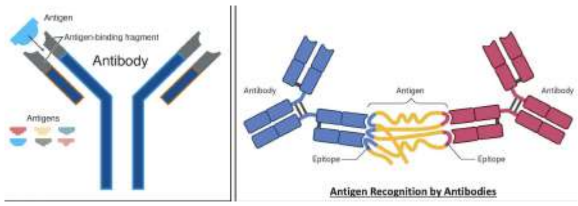

Figure 16: Immunoassay Principle

19. Immunoassay Method 1 : Enzyme-linked immunosorbent assay (ELISA)

- ELISA = Enzyme Immunoassay (EIA)
- A Plate-based Assay Technique

- Used for Detecting + quantifying Soluble Substances
  ⇒ E.g. Peptides, Protein, Antibodies, Hormones
- Type of ELISA : Direct ELISA, Indirect ELISA, Sandwich ELISA, Competitive ELISA

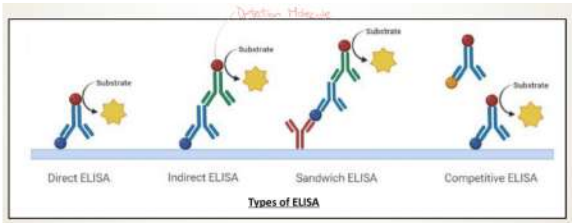

Figure 17: ELISA

- Stopping Signal : Turning to Yellow (TMB) / Green-Blue (ABTS)
- Non-stopping Signal : blue (TMB & ABTS)
- Antibody is directly conjugated to Enzyme Horseradish Peroxidase (HRP) substance
  ⇒ E.g. TMB, ABTS
- Coloured Products are produced and measured
  ⇒ Use Specialized Plate Reader to look at the colour intensity of the solution

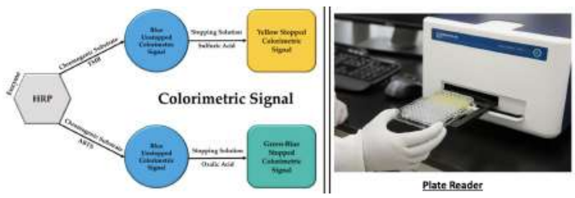

Figure 18: Plate Reader

20. Immunoassay Method 2 : Radioimmunoassay (RIA)

- Radioisotopes : Conjugated with Antigens / Antibodies
- Classic RIA : Based on Competitive Binding
- Unlabelled Antigen competes with Radiolabelled Antigen
  ⇒ Used to bind to an antibody

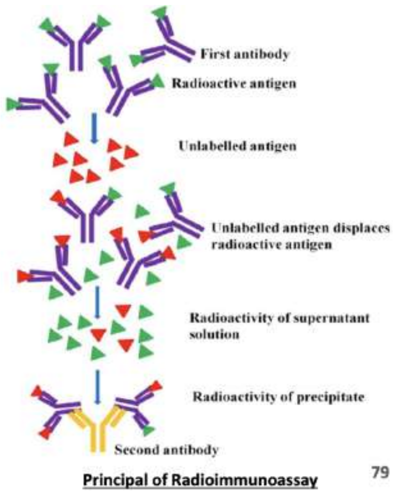

Figure 19: Principle of Radioimmunoassay

- Unlabelled Antigens : Replace Labelled Antigens which are already linked with the antibodies
  ⇒ It can increase the amount of Free Radiolabelled Antigens in the solution
- Concentration of Free Labelled Antigens is Directly Proportional to The Bound Unlabelled Antigens
- Antigen Concentration Unit : nmol/L

21. Immunoassay Method 3 : Direct Fluorescent Antibody (DFA)

- A rapid Microscopic Procedure
- Function 1 : Can detect the presence of a particular antigen
  ⇒ Use Fluorescently Labelled Antibodies
- Function 2 : Visualize certain bacteria and viruses that are Difficult to Isolate and Culture from Patient Sample

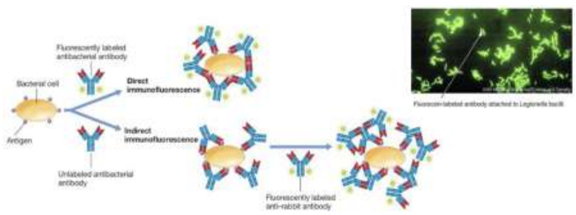

Figure 20: Direct Fluorescent Antibody (DFA)

2 Lecture 5 Spores and their Significance

1. Spores
   (a) Survival : Can survive the initial cooking / boiling
   $\Rightarrow$ If no heat, heat can kill all vegetative bacteria
   (b) Activation : The heat of cooking activated the spores
   (c) Germination : Cool down time too long at room temperature
   $\Rightarrow$ Spore Germinated + Bacteria Grew
   (d) Inadequate Stir-frying (Heat is not uniform)
   $\Rightarrow$ High Bacteria Count
2. Endospore
   - Highly Resistant
   - Dormant Cells : Not activated
     $\Rightarrow$ Produced by certain Gram-positive
     $\Rightarrow$ Not all bacteria can form spores
   - Spores formed within vegetative cells
     $\Rightarrow$ Released from the mother cell at the end of Sporulation
3. Endospores Characteristics
   - Resistant to Environmental Stresses
     $\Rightarrow$ E.g. Heat, Drying, UV Radiation, Chemical Disinfectants
   - Metabolically Inactive but still Alive
   - Survival but not Reproductive
     $\Rightarrow$ 1 Cell $\rightarrow$ 1 Spore
   - Form a vegetative cell in a suitable environment
     $\Rightarrow$ 1 Spore $\rightarrow$ 1 Cell
4. Spore-Forming Bacteria
   - Some bacteria form Endospores as part of their life cycle
     $\Rightarrow$ Formation of Endospores : Escape Gradual Death
   - Spore-forming Bacteria

- A major threat in Heat-Treated Food
- Spores (Endospore) are Heat Resistance
  ⇒ Survive Heat Treatment
  ⇒ E.g. Cooking at 100°C

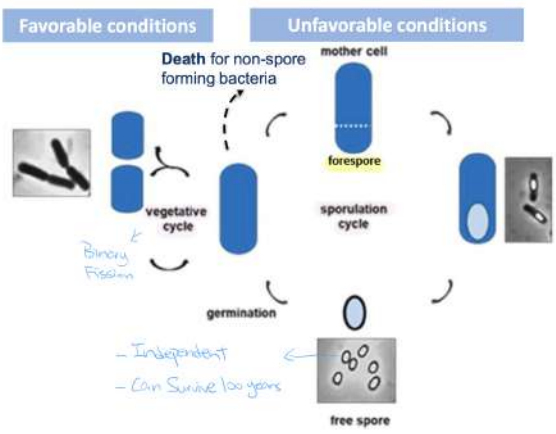

Figure 21: Spore-Forming Bacteria

5. General Features of Spore-Forming Bacteria
   - Most are Soil Bacteria
   - Found in Food Contaminated with Soil
     ⇒ E.g Vegetables, Grains
   - Exception : ***Clostridium perfringens*** is also a Spore-Forming Bacteria
     ⇒ Found in Animal Intestine
     ⇒ Present in both meat and soil-contaminated food
   - Relatively Poor Competitors in Raw Food
     ⇒ Do not grow in food
   - If competitors cannot grow (Cook→ Competitor Bacteria Die) → Spore can use all the nutrient without competition → Able to grow
6. Important Genera Spore-forming Bacteria
   (a) ***Clostridium* (Anaerobic)**

- ***C. botulinum*** : Deadly Neurotoxin in canned foods (Botulism)
- ***C perfringens*** : Toxico-infection in meats / gravies

(b) ***Bacillus* (Aerobic / Facultative Anaerobic)**

- ***B. cereus*** : Emetic Toxin (Rice) and Diarrheal Toxin
  $\Rightarrow$ A Pathogen
- ***B. subtilis*** : Spoilage Organism in hest-processed food

7. Food Industry Implication : Canning is designed around killing ***C. botulinum*** Spores

8. Structure of Spores : Different from Vegetative Cells

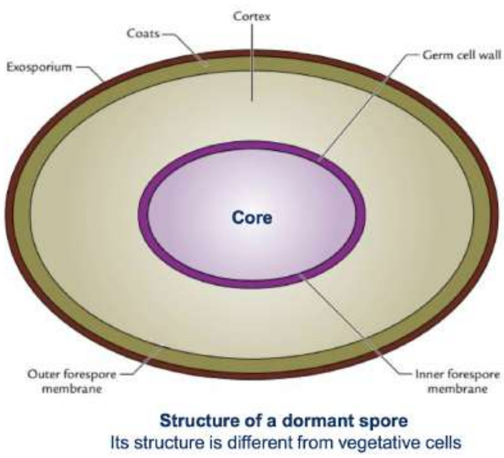

Figure 22: Spore Structure

(a) Exosporium

- Thin
- Protein-rich Layer
- Not always Presents
- Protect the Spore
- Influences spore surface adhesion

(b) Spore Coat

- Think
- Multilayered Coat

- Made by Protein
- Protect from chemicals and lytic enzymes
  ⇒ Lytic Enzyme : Lysozyme
  ⇒ Can degrade bacteria cell wall → Kill Bacteria

(c) Outer Forespore Membrane (Outer Membrane)

- Functional Membrane in Developing Forespore
- Serve as a barrier between Mother Cell & Developing Spore

(d) Cortex

- Formed by Think Peptidoglycan (PG) Layer
  ⇒ 50% of Spore Volume
- Maintaining Core **Dehydration**
  ⇒ Heat Resistance

(e) Germ Cell Wall

- Formed by Thin Peptidoglycan (PG) Layer Beneath Cortex
- Similar Structure to Vegetative Cells
  ⇒ After Germination → Become Vegetative Cell Wall

(f) Inner Forespore Membrane (Inner Membrane)

- Phospholipid Content is similar to Vegetative Cells
- After Germination → Become Vegetative Cell Membrane
- Functional Membrane : Very Strong Permeability Barrier
  ⇒ Almost all molecules cannot enter
  ⇒ E.g. Water

(g) Core

- Contains DNA, Ribosomes, Enzymes
- Unique Chemical Compound
  - Ca-DPA (Dipicolinic Acid) : Ca^{2+} Chelated with DPA
    ⇒ Stabilizs Spore DNA
  - SASPs (Small Acid-Soluble Protein)
    ⇒ Bind tightly to spore DNA
    ⇒ Stabilize Spore DNA

- Gel-like Dehydrated State (Dormancy)
  - Very Low Water Content
    $\Rightarrow$ 10 times lower than vegetative cells
  - No Metabolism
  - Responsible for Spore Resistance & Dormancy
- The Unique Chemical Compound is contributed to Heat Resistance

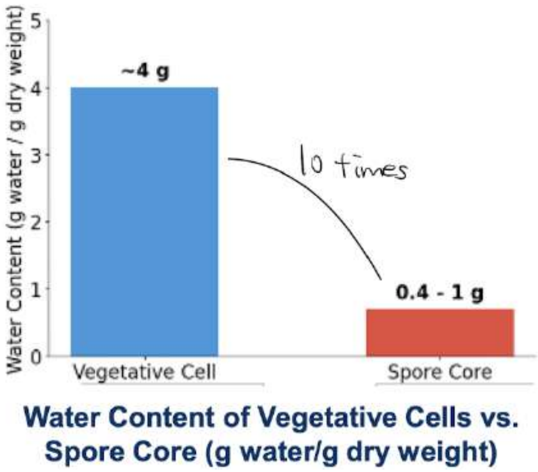

Figure 23: Water Content of Spore Core

9. Core Composition

   (a) Macromolecules

      i. Spore DNA = Vegetative Cell DNA
         $\Rightarrow$ But Different Structure
         $\because$ SASPs Binding

      ii. Ribosomes : Present but inactive
         $\therefore$ No Metabolisms & Protein Synthesis

      iii. Small Acid-Soluble Protein (SASPs)
         - Unique to Spores
           $\Rightarrow$ 10% – 20% of Total Protein
         - Important to Spore Resistance
         - Bind tightly to DNA
           $\Rightarrow$ Change DNA Conformation to a compact shape
           $\Rightarrow$ Physically Wrap Up

- Protect against UV Radiation, Chemicals, Heat, Desiccation

iv. Enzymes
- No Biosynthetic Enzymes
  $\therefore$ No Growth
- Have Catabolic Enzymes
  $\Rightarrow$ Ready for Rapid Energy Generation
  $\Rightarrow$ Will Active when germination starts

(b) Small Molecules

i. Large Amount of Ca-DPA (Calcium Dipicolinate)
- Unique to Spores
- 5% – 15% Spore Dry Weight
- Displaces Water in the Core $\rightarrow$ Core Dehydration
- Stabilizes the DNA and Protein $\rightarrow$ Heat Resistance

ii. Acidity
- Core pH : 6.5
  $\Rightarrow$ Vegetative Cells pH : pH 7.5 – 8
- Acidic Environment $\rightarrow$ Enzymes Inactive

iii. Low levels / Absence of High Energy Compounds
- No ATP & NADH
  $\Rightarrow$ No energy for Cellular Processes

iv. Water
- Very Low Water Content
- Ions & Proteins are immobile
- No Metabolic Activity & No Protein Unfolding

10. Spore Dormancy

- Spores are Metabolically Dormant
  - No detectable metabolism of endogenous or exogenous compounds
  - E.g. Spore core contains enzyme-substrate pairs that remain stable for months to years
- Cause of Dormancy : Core Dehydration

(a) Low Water Content of Spore Core
$\Rightarrow$ Prevent Protein Mobility and Enzyme Action

(b) Present Enzymes and Substrates but Cannot Interact
$\Rightarrow$ Only react when water enters after germination

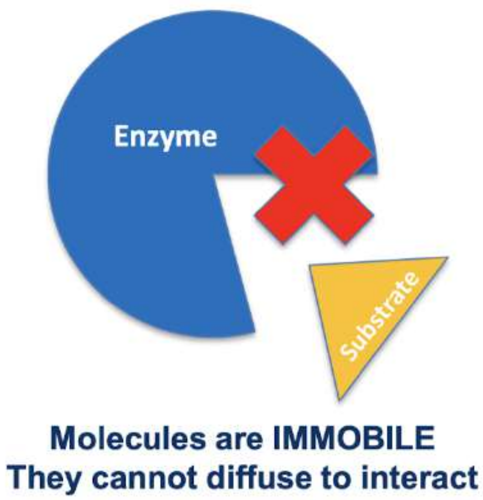

Figure 24: Spore Dormancy

11. Spore Resistance
   (a) Heat Resistance (Most Important for Food)
      - Many Spores can withstand 100°C (Moist Heat) for several minutes
        $\Rightarrow$ Moist Heat = Boiling
        $\Rightarrow$ Denature Protein
      - Target : Moist Heat damages Proteins
        $\Rightarrow$ Not Damage DNA
      - Mechanism 1 : Core Dehydration $\rightarrow$ Prevent Protein Denaturation
        $\Rightarrow$ Ca-DPA remove water in core
      - Mechanism 2 : Cortex $\rightarrow$ Maintains Core Dehydration
   (b) Radiation Resistance
      - Resistant to Gamma Radiation and UV Radiation

- Target : Radiation damage DNA
- Mechanism : Binding of SASPs → DNA

(c) Pressure

- Resistant to high pressure
- Target : Pressure disrupts Inner Membrane Integrity
- Ca-DPA : Play a Role in Pressure Resistance

(d) Freezing & Desiccation Resistance

- Spores withstand multiple cycles of freeze-drying
- Target : Involves DNA damage
- Mechanism 1 : Binding of SASPs → Prevents DNA damage
- Mechanism 2 : Ca-DPA in the core → Provides Additional Desiccation Resistance

(e) Chemical Resistance

- Resistant to Chemicals (Sanitizer), pH Extremes, and Lytic Enzymes
- Target : Depending on the chemicals
  ⇒ May include Inner Membrane, DNA, Cortex
- Mechanism 1 : Spore Coat → Provide Initial Barrier
- Mechanism 2 : Low Permeability of Inner Membrane → Prevent most molecules entering the core
- Mechanism 3 : Binding of SASPs to DNA → Prevent DNA Damage by Molecules that Enter the Core

12. Lifecycle of Endospores

(a) Sporulation

- Conversion of Vegetative Cell to Dormant Spores
- Triggered by Starvation

(b) Germination

- Conversion of Spore to Vegetative Cell

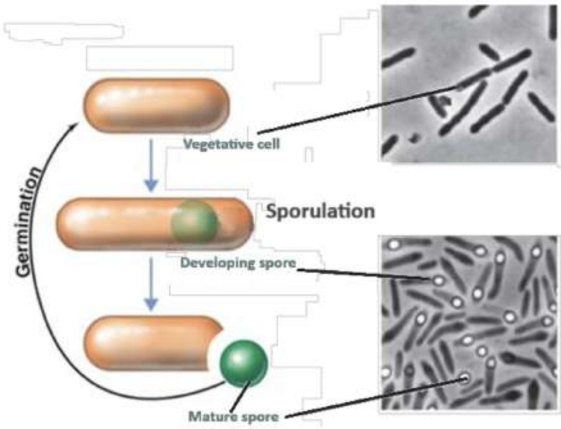

Figure 25: Endospore Lifecycle

13. Sporulation
    - Survival Strategy under **Harsh Environmental Condition**
    - Spores are made in response to Environmental Stress / Nutrient Depletion
    - Example of Environmental Stress : Change in Temperature, Water Activity (A_W), pH
    - Reduction in nutrient availability
      ⇒ E.g. Carbon, Nitrogen, Phosphate Source
    - Time Required : Several Hours
    - Contain 7 stages
      ⇒ Based on Morphological Characteristic of Cells
14. Sporulation Stages
    (a) Stage I : Axial Filamentation of Formation
    - Cell contains Two Copies of Chromosomes
    - Alignment of Chromosomes in an Axial Filament
      ⇒ Stretching one pole of the cell to another side

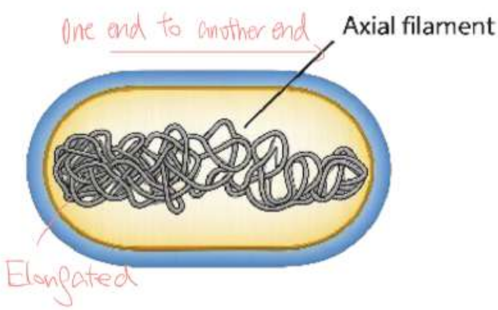

Figure 26: Stage I : Axial Filamentation of Formation

(b) Stage II : Asymmetric Septation
   - Unequal Cell Division
   - Cell Membrane forms Asymmetric Septum
     ⇒ Divided the Sporulating cell into Mother Cell & Forespore Compartments
   - Each Compartment contain One Copy of Chromosome

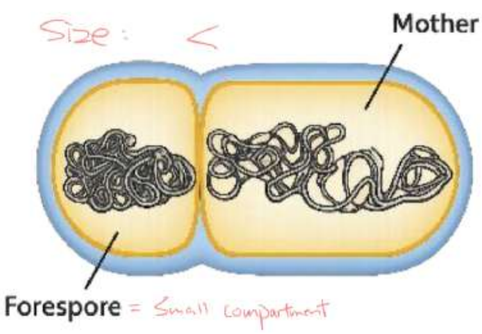

Figure 27: Stage II : Asymmetric Septation

(c) Stage III : Engulfment
   - Mother Cell Plasma Membrane Grows
   - Engulfs the Forespore
   - Surrounding the forespore with complete membranes
     ⇒ Forespores (FS) has two Membranes : Inner FS Membrane & Outer FS Membrane
   - FS Synthesize SASPs : During Stage III - Stage IV Transition

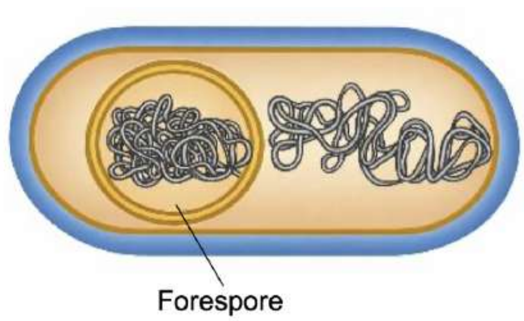

Figure 28: Stage III : Engulfment

(d) Stage IV : Cortex Formation
- Think Cortex is formed with Peptidoglycan (PG) between Inner & Outer FS Membranes
- Cortex is critical for **Spore Heat Resistance and Dormancy**
  $\Rightarrow$ Cell Dehydration
- Germ Cell Wall : Formed with Peptidoglycan (PG) at the same time
- Synthesis of Exosporium Occurs

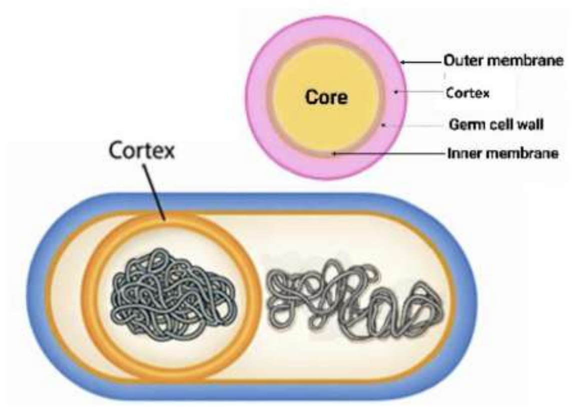

Figure 29: Stage IV : Cortex Formation

(e) Stage V : Coat Deposition
- Proteinaceous Coat is formed outside the outer FS membrane
- MC produces a large amount of DPA $\rightarrow$ DPA is trans-

ported into the developing FS in exchange for water $\rightarrow$ Complex with Ca^{2+} $\rightarrow$ Dehydrate Forespore
$\Rightarrow$ During Stage V-Stage VI transition
$\Rightarrow$ This means Synthesis DPA and Pump into Forespores

- Contributing to Spores Dehydration

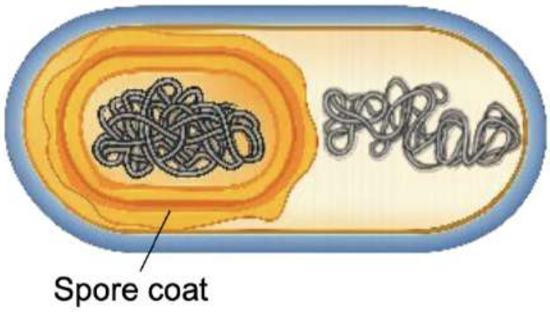

Figure 30: Stage V : Coat Deposition

(f) Stage VI : Spore Maturation

- Spore core undergoes Final process of Dehydration
- Spore becomes Metabolically Dormant & Resistant to Heat
  $\Rightarrow$ Cannot be killed by heat = Dehydration

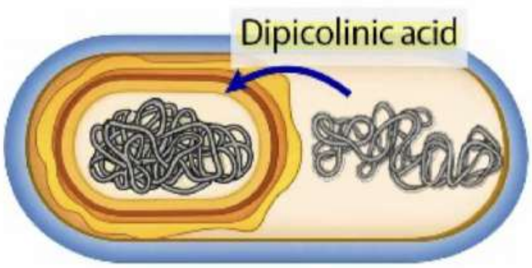

Figure 31: Stage VI : Spore Maturation

(g) Stage VII : Release of Spores

- Enzymatic Lysis (Break) the Mother Cell $\rightarrow$ Release the spore

Figure 32: Stage VII : Release of Spores

15. Germination
    - Dormant Spores germinate to become Vegetative Cells
      ⇒ When they Germinate, Dormant Spores Resume Metabolism
    - Transition from Spores to Vegetative Cells : Three Phase
      ⇒ Activation, Germination, Outgrowth
    - Germination only occurs when there is favorable condition
      ⇒ Mature spore stay at mature spore state if there is worse condition = No Germination

Figure 33: Germination

16. Spore vs Vegetative Spore Transition Stages
    (a) Stage I : Activation
    - A process that allow spores to undergo Germination
    - Most can germinate without activation
    - But Activation can ↑ Rate of Germination + ↑ Completeness (More Spores in a population germinate)
    - Activation Methods
      - Use Sublethal Heat Treatment : 10 minutes with 80°C
      - Low pH
      - High Pressure

- Can add some Chemical
- Reversible Process in some species
  $\Rightarrow$ Reverse when nutrition is not enough
  $\Rightarrow$ Spores return to dormancy
- Heat (Cooking) often activates spores $\rightarrow$ Germinate more rapidly and completely in food

(b) Stage II : Germination

- Active Metabolizing Germinated Spore : Release DPA + Degrade Cortex and SASPs $\rightarrow$ Germinated Spores
- Triggered by a **Germinant (It induce Germination)**
  - Nutrient Germinant : Amino Acid, Sugars
  - Non-Nutrients Germinant : Salt, Ca-DPA, Peptidoglycan (PG) Fragments
- Germinant Metabolism is not needed
- Nutrient Germinant interacts with specific receptor
- Nutrient Germinant binds to receptor Protein in Inner Membrane

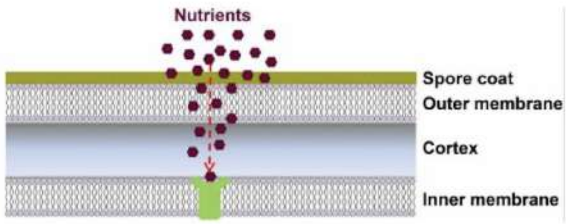

Figure 34: Nutrient Entering

- Events During Germination
  i. Release of Protons & Cations
     - Spore reach a point of commitment
     - Germination Process become irreversible
       $\Rightarrow$ Even germinants are removed
  ii. Release of CaDPA $\rightarrow$ Spore loses 30% of its dry weight
  iii. Core Hydration : Water Replaces released DPA and Ions

iv. Cortex Degradation : Removing the Restrictive Cortex $\rightarrow$ More Water Uptake $\rightarrow$ Rapid Increase in Spore Core Volume

v. SASPs Degradation : Breakdown SASPs $\rightarrow$ Produced Amino Acids $\rightarrow$ Protein Synthesis Begins

- Germination Completion : Loss of Most Resistance
$\Rightarrow$ Loss of Spore Cortex & Hydration & Swelling of the core in Germinated Spores

(c) Stage III : Outgrowth

- Germinated Spore transitions into Actively Growing Vegetative Cells
- Requires 25 minutes after the start of germination until first cell division
- Events during outgrowth

i. RNA Synthesis : Use Stored Nucleotides / Degraded Spore RNA

ii. Protein Synthesis
  - Use amino acids from SASP degradation in the beginning
  - Exogenous nutrients required to complete outgrowth
  $\Rightarrow$ E.g. Vitamins, Minerals, Ketones

iii. Membrane and Cell Wall Component Synthesis $\rightarrow \uparrow$ Volume of Outgrowing Spore

iv. DNA Repair : Repair of DNA damage during dormancy

v. DNA replication late in outgrowth

- New Vegetative Cell burst out of the spore coat

17. Controlling Spores in Food Products

(a) Acidification

- Goal : Prevent Spore Germination
$\Rightarrow$ Spores may survive but would not germinate / grow to produce toxin anymore

- Most pathogenic endospores cannot germinate below pH 4.6
- Food is still save even spore exist
  $\Rightarrow$ Ensure the spore is not germinated
- Example : Fruit jam (pH 3) by adding citric acid

(b) Thermal processing
- Goal : Kill Spores
- Example : Retorting ($> 121°$C for 3 minutes) for low-acid canned food (pH > 4.6)
  $\Rightarrow$ Kill ***C. botulinum*** spores

(c) Chemical Preservative
- Goal : Prevent Spore Germination
- Example : Nitrites (In cured meats) : Inhibit ***C. botulinum*** Spore Germination

18. Control of Spore-Former Foodborne Disease

(a) Intoxication
- The majority of spore-former pathogens form toxin in food
- Bacteria must grow extensively in food to cause disease
- Soil contamination is an important factor

(b) Soil Contamination : All foods harvested from the soil contain at least a few spores

(c) Cooking promotes microbial growth of spore-formers

(d) Heat Activation
- Eliminating Competing Bacteria
- Triggering Dormant Spore to Germinate

(e) Proper Storage of Cooked Food is the key to control
- Cooking induces Germination but growth occurs
  $\Rightarrow$ When food stored under a favorable conditions
- Food should be Refrigerated / Kept it very hot

3 Lecture 6 Beneficial Microorganisms in Foods

1. 3% of microbe species in adults guts are also found in foods
   (a) Beneficial Microorganisms
      - Probiotics
      - Starters for Food Fermentation
         - Inhibit Spoilage / Pathogenic Microbes
         - $\uparrow$ Shell-life of Food
         - Enzymatic Activity

   (b) Microorganisms used to produce specific compounds
      - Bioactive Compounds
        $\Rightarrow$ E.g. Vitamins, Biopeptides, Antioxidants
      - Enzymes

   (c) Spoilage
   (d) Pathogenic Microorganisms
2. Microorganisms in Food Application & Food Production
   - Tradition Fermentation : Alcohol Production, Food Preservation, Taste and Texture Development
   - Enhanced Fermentation : Strain Chosen
     $\Rightarrow$ Used to improve Taste, Texture, Health Benefits
   - Animal Feed : Nutritional Supplement, Primary Food Source, Nutraceutical and Therapeutic Platform
   - Ingredient Production : Production and Purification of native and recombinant food
   - Meat & Diary Alternatives : Microbial meat $\rightarrow$ Recreating Nutrition, Taste and Texture
   - Single Cell Protein : Primary Source of Protein, Complete food source, Health Supplement
3. Meat & Diary Alternatives + Single Cell Protein
   - Mycoprotein = A protein-rich food ingredient from Fungal Mycelium

- Want to Mimic Animal Protein

4. Beneficial Intake of Microorganism in food

   - Early Human Diet : High Microbial Numbers
     $\Rightarrow$ High levels of Interaction with Immune System
     $\Rightarrow$ Health Status Increased
     $\because$ Compensate the interaction of immune system and microbes
   - Industrialized Human Diet : Low Microbial Numbers
     $\Rightarrow$ Low levels of Interaction with Immune System
     $\Rightarrow$ Immune System will decrease the efficiency through generation
     $\Rightarrow$ Health Status Decreased
   - Microbe Supplemented Diet : Added Microbial Numbers
     $\Rightarrow$ High levels of Interaction with Immune System
     $\Rightarrow$ Health Status Increased

5. Biotic Family

   - Probiotics : Provide Health Benefits
   - Prebiotics : Supplement of Food
   - Synbiotics : Probiotics + Prebiotics
   - Postbiotics : Used to inactivate Bacteria
     $\Rightarrow$ Provide Health Benefits
   - Fermented Foods : Controlled Microbial Growth
   - Advanced Microbiome Therapeutics

6. Probiotics

   (a) Definition : **Living Microorganisms** that, when administered in **adequate amount** confer a **health benefit** on the host

   (b) Microorganism = Physical Term
   $\Rightarrow$ E.g. Fungi, Animals, Virus, Bacteria
   $\Rightarrow$ Viruses is not alive

   (c) Living = Biological Term

(d) Adequate Amount : Cannot too much and too little

(e) Estimated Range of Adequate amount : $10^7 - 10^9$ living cells of a strain/day

(f) Health Benefits : Probiotics ⇔ Gut Microbiota (Intestinal)

7. Health Benefits of Probiotics

(a) Widespread : Observed across Taxonomic groups

(b) Frequent : Species-level effects

(c) Rare : Strain-specific effects (Genetic Variant)
⇒ Example 1 : ***Lactobacillus (Lacticaseibacillus) rhamnosus*** GG : Reduces Diarrhea Duration
⇒ Example 2 : ***Bifidobacterium lactis*** HN019 : Enhances Immune Response

(d) Short-chain Fatty Acid

- Butyrate
  - Primary source of energy for Colonocytes
  - Potential Anticancer Activity
    - Tumor Cells : Apoptosis Induction
    - Modulate Gene Expression
  - Intestinal Gluconeogenesis Activator : Activated by **Butyrate**
    ⇒Function : Improve Glucose Homeostasis + Increase Energy Expenditure through gut and brain
- Propionate
  - Energy source for the Colonocytes
  - Gluconeogenesis Modulator : Activated by Propionate
  - Get from Liver Satiety regulator
  - Also get from Lypogenesis
- Acetate
  - Cholesterol Metabolism
  - Lipogenesis
  - Regulation of Appetite
    ⇒ Activated by Acetate

8. Fermented Food

- Type of Raw Material of Origin
  (a) Cereals and Seeds → E.g. Breads
  (b) Milk & Tea
  (c) Legumes
  (d) Fruits, Vegetables & Tubers
  (e) Meat
  (f) Fish & Shellfish

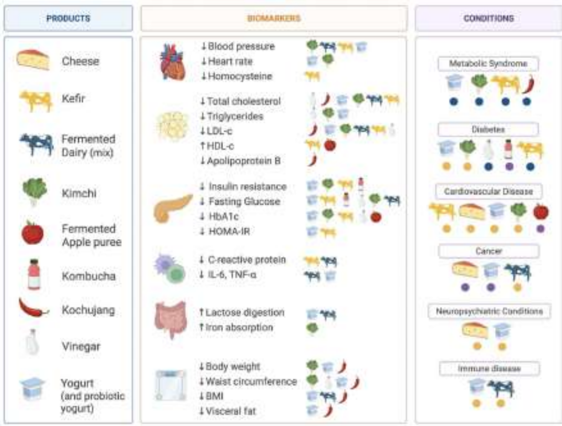

Figure 35: Fermented Foods & Human Health

- Type of Fermentation
  (a) Alcoholic : Ethanol & Gas, Yeasts
  ⇒ E.g. Breads
  (b) Acetic : Acetic Acid, Acetobacter (Bacteria)
  ⇒ E.g. Vinger
  (c) Lactic : Lactic Acid (Fatty Acid)
  (d) Butiric : Butiric Acid Clostridium (Non-beneficial Bacteria)
  (e) Propionic : Propionic Acid, ***Propionibacterium***
  ⇒ E.g. Sweet Cheese
  (f) Alkaline : Ammonia, ***Bacillus***, Mold
- Reason for Food Fermentation

(a) Enrichment of the diet (Flavor Aromas, Textures)
$\Rightarrow$ E.g. Milk
(b) Preservation of Food
$\Rightarrow$ E.g. Lactic Acid, Alcoholic, Acetic Acid, Alkaline Fermentation, Antimicrobial Compounds
(c) Biological Enrichment of Food Substances : Protein, Essential Amino Acid, Essential Fatty Acid, Vitamins
$\Rightarrow$ Increase food nutritional properties
(d) Detoxification during food fermentation processing
$\Rightarrow$ Reduce the presence of antinutritive and toxic compounds
(e) $\downarrow$ Cooking Times & Fuel Requirement
(f) $\downarrow$ Production Costs and Loss from Food Spoilage
$\because$ Have Controlled & Efficient Fermentation
(g) $\uparrow$ Nutrient Content of Food

9. Lactic Acid Fermentation

- Two types of Lactic Acid Fermentation : Homofermentative, Heterofermentative
- Gram-positive Bacteria : Use carbohydrates for the only carbon source & Provide Energy
- Phylum of Gram-positive Bacteria : **Bacillota** (Firmicutes)
- Vegetable-based Fermentation
  - Fermentate the Vegetable
  - Salt is crucial
  - Spoilage through Putrefaction & Softening
  - Become Suan-Tsai
  - LAB : ***Lactobacillus, Leuconostoc*** and ***Pediococcus***
- Soybean-based Fermentation
  - Bacillus $\rightarrow$ Alkaline Fermentation (Ammonia) $\rightarrow$ Smelly
  - From Lactic Acid Fermentation to Alkaline Fermentation
  - LAB : ***Enterococcus, Lactobacillus, Lactococcus, Leuconostoc, Pediococcus, Streptococcus, Tetragenococcus*** and ***Weissella***

- Example : Stinky Tofu

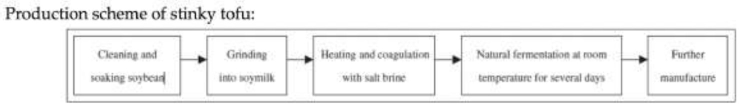

Figure 36: Stinky Tofu Fermentation

- Meat-based Fermentation
  - LAB : ***Lactobacillus, Leuconostoc, Weissella*** and ***Pediococcus***

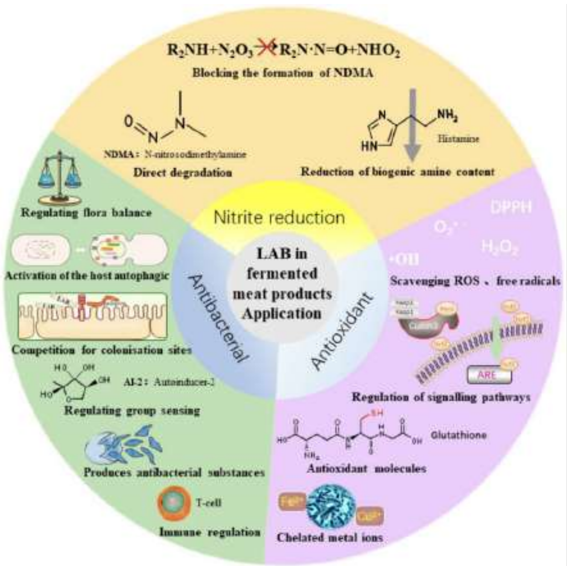

Figure 37: Meat-based Fermentation

10. Applications of LAB in Food Industry
    (a) Degradation of Macromolecules
    - Degradation of indigestible disaccharides polysaccharides
    - Lactose, Starch, Cellulose, Fructan, Hemicellulose → Monosaccharides, Organic Acids
    - Degradation of Mycotoxins

    (b) Degradation of Protein and Catabolism of Amino Acids
    - Reducing the allergenicity
    - Improve the digestibility of protein in food

- Enhance nutritional value
- Influence flavor

(c) Synthesis of Specific Molecules
- Extend food preservation duration
- Natural and Safe Preservative in food products
- Example : Bacteriocins, Vitamins

(d) Metabolic Engineering Strategies Application in LAB

11. Fungal Fermentation of Foods
    - Yeast Fermentation : Bread, Tea, Soy Sauce
    - Produce bioactive compounds
      ⇒ E.g. Bioactive peptides, Chitin, Chiosan, Ergosterol

12. Risks of Using Microorganisms in Industry
    - Antimicrobial Resistance
    - Production of Biogenic Amines
    - Production of Mycotoxins

13. Future Direction

(a) Health Impact
- Improve Nutritional Value
- Increase Gut Microbiome Diversity
- Postbiotics : Used to Inactive Bacteria
- Bioactive Natural Compounds
- Reduction of Hunger

(b) Environmental Impact
- Increase Biodiversity
- Decreased Land Use
- Decreased Pollution
- Decreased Carbon Emission
- Increase Resilience of Food System

(c) Production of Microbial Foods

- Dry Weight Values of Solein : Protein Content (78%) + Fat (6%) + Carbohydrates (2%) + Total Dietary Fibres (10%) + Minerals (4%)

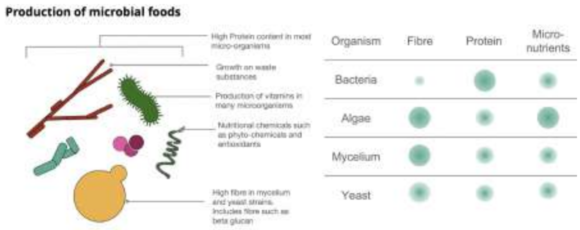

Figure 38: Production of microbial foods

(d) Advance Microbiome Therapeutics